RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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Academic year 2024-2025 (Odd Sem)

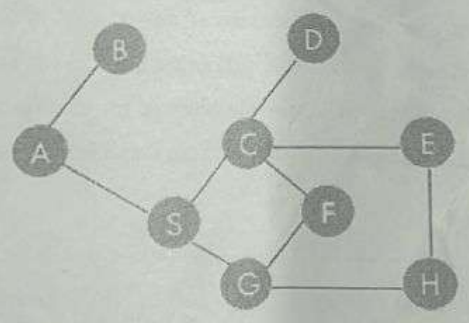
DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

Date	08-01-2025	Maximum Marks	10+50
Course Code	IS353IA	Duration	2 hours
Sem	V	CIE -II	
UG/PG	UG	Faculty: AS/VH/VG/JS/SHRS/ARA/MEM	

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

(Common to CSE/ISE/CD/CY)

Q. No.	QUIZ	M	BT	CO
1	The process of removing details from a representation is called _____ .	1	1	4
2	The depth of a tree is given by "d". While using Iterative deepening search, the children of root are generated _____ times.	1	3	2
3	Suggest a method to characterise the quality of a heuristic	2	1	5
4	The k-NN algorithm is sensitive to the choice of K. Recommend a way to reduce it	1	2	1
5	The Naïve Bayes classifier is robust to irrelevant attributes. Give reason	2	3	4
6	Suggest 4 different ways to evaluate the performance of an algorithm	2	3	5
7	A node in a tree is represented by 4 components . Which are they ?	2	2	4

Q. No.	TEST	M	BT	CO
1	Apply BFS and DFS to the following graph and compare the results . Show all the necessary steps during the traversal and reasons  Also discuss the pros and cons of BFS and DFS	10 (6+4)	3	1
2	Find the most cost-effective path to reach from start state A to final state J using A* Algorithm for the graph in Figure :1	10	4	2

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	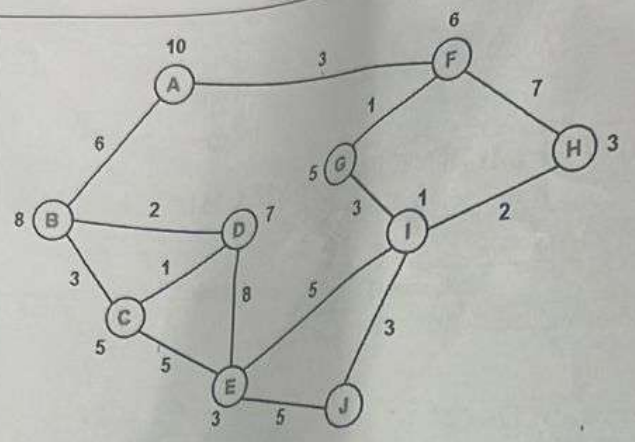 Figure 1			
3	Consider the statements given below . State whether you agree or disagree with the statements . Give valid reasons for each **a**. Depth-first search always expands at least as many nodes as A* search with an admissible heuristic. **b**. h(n) = 0 is an admissible heuristic for the 8-puzzle. **c**. A* is of no use in robotics because percepts, states, and actions are continuous. **d**. Breadth-first search is complete even if zero step costs are allowed. **e**. Assume that a rook can move on a chessboard any number of squares in a straight line, vertically or horizontally, but cannot jump over other pieces. Manhattan distance is an admissible heuristic for the problem of moving the rook from square A to square B in the smallest number of moves.	10	3	5
4	Consider the data given in Table 1	10	4	2

Instance	A	B	C	Class
1	0	0	0	+
2	0	0	1	−
3	0	1	1	−
4	0	1	1	−
5	0	0	1	+
6	1	0	1	+
7	1	0	1	−
8	1	0	1	−
9	1	1	1	+
10	1	0	1	+

Table 1

From the data given,

(a) Estimate the conditional probabilities for P(A|+), P(B|+), P(C|+), P(A|−), P(B|−), and P(C|−).

(b) Use the estimate of conditional probabilities given in the previous question to predict the class label for a test sample (A = 0,B = 1, C = 0) using the Naive Bayes approach.

(c) Estimate the conditional probabilities using the m-estimate approach, with p = 1/2 and m = 4.

(d) Repeat part (b) using the conditional probabilities given in part (c).

5 Consider the dataset given below

Using K-NN Algorithm, classify the unknown instance : brightness =20 and saturation =25 to appropriate class . Assume k=5

Brightness	Saturation	Class
40	20	Red
50	50	Blue
60	90	Blue
10	25	Red
70	70	Blue
60	10	Red
25	80	Blue

Also give any 5 characteristics of K-NN algorithm

5+5 | 3 | 5